

Mirofish Comprehensive Sports Betting Intelligence Report

Version: 1.0 | **Date:** 2026-04-02

Audience: Mirofish prediction and execution engine

Objective: Build a robust, risk-aware, high-performance sports/event trading framework that behaves like a hybrid of elite quantitative bettor, Las Vegas high-stakes risk manager, and exchange microstructure trader.

1) Executive Objective

Mirofish must optimize for **long-run risk-adjusted edge**, not short-run hit rate.

1. Generate calibrated probabilities (not just picks).
2. Convert every opportunity into fee/vig-adjusted EV.
3. Execute only when edge survives friction (limits, liquidity, slippage, fees).
4. Protect bankroll through strict risk controls and correlation-aware sizing.
5. Continuously learn from CLV, realized PnL, and model calibration drift.

2) Legal and Operational Guardrails

- Operate only in legally permitted jurisdictions and products.
- Respect book limits, market terms, and prohibited use clauses.
- Keep full audit trails for data source, timestamp, model version, and execution rationale.
- Treat all outputs as probabilistic recommendations, never guarantees.

3) US Market Context

Post-2018 structure

The US market is fragmented by state regulation and operator footprint. This fragmentation creates opportunity through line shopping, timing mismatches across books, stale derivatives/props, and varying limit/risk tolerances.

Common books (state-dependent)

- FanDuel
- DraftKings
- BetMGM
- Caesars
- ESPN BET
- Fanatics

- BetRivers
- bet365 (state dependent)
- Hard Rock Bet (state dependent)

Exchange/event-contract venue

- Kalshi (binary YES/NO contracts, CFTC-regulated model)

4) Kalshi / Event-Contract Intelligence

Core mechanics

- Binary contracts settle to **\$1 or \$0**.
- Contract price (for example 0.62) approximates market implied probability before costs.
- YES/NO are tradable before settlement; order-book quality matters.

Mirofish EV treatment

$$\text{EV}_{\text{yes}} = p_{\text{model}} - c - f$$

$$\text{EV}_{\text{no}} = (1 - p_{\text{model}}) - (1 - c) - f$$

$$\text{Break-even YES probability} = c + f$$

Always fee-normalize and account for spread, depth, and slippage. A strong model probability can still be a poor trade after execution costs.

5) Core Sportsbook Math Standards

American odds to implied probability

$$\text{If } +A: p_{\text{raw}} = 100 / (A + 100)$$

$$\text{If } -A: p_{\text{raw}} = |A| / (|A| + 100)$$

No-vig conversion (2-way markets)

$$p1_{\text{fair}} = p1_{\text{raw}} / (p1_{\text{raw}} + p2_{\text{raw}})$$

$$p2_{\text{fair}} = p2_{\text{raw}} / (p1_{\text{raw}} + p2_{\text{raw}})$$

Expected value

$$EV_per_ \$1 = p * (d - 1) - (1 - p)$$

Only enter when EV remains positive after fees, slippage, and uncertainty adjustments.

6) Professional Trading Mindset

As an elite bettor

- Demand quantified edge for every entry.
- Pass when edge is unclear.
- Prefer repeatable small edges over volatile swings.
- Track expected vs realized outcomes separately.
- Prioritize CLV as a process metric.

As a Vegas high-stakes risk manager

- Track liability concentration by side/game/league/day.
- Manage correlation risk (same-team, same-game, derivative chains).
- Adapt limits by confidence, liquidity, and market stability.
- Reprice aggressively during injuries/starter/weather shocks.
- Differentiate true pricing from balancing/liability pricing.

As an exchange microstructure trader

- Incorporate spread, depth, queue position, and fill probability.
- Prefer limit-style execution where practical.
- Treat execution quality as part of model quality.

7) MLB Modern History and Updated Baselines

Era summary

1. Late 1990s-2000s: offense-heavy period.
2. Moneyball era: OBP and efficiency optimization.
3. Statcast era: EV, launch angle, pitch design, matchup granularity.
4. 2019: peak HR environment.
5. 2023 rule changes: pitch clock, larger bases, shift limits altered run environment and pacing.
6. 2024-2025: environment stabilized relative to 2023 spike.

2025 MLB league averages (per team game)

Metric	Value
Runs/Game	4.45
Batting Avg (BA)	.245
OBP	.315
SLG	.404
OPS	.719
HR/Game	1.16
BB/Game	3.16
SO/Game	8.36
SB/Game	0.71

2025 pitching baseline

Metric	Value
ERA	4.16
FIP	4.16
WHIP	1.289
K/9	8.5
BB/9	3.2
HR/9	1.2

MLB implications for Mirofish

- Model starter quality + bullpen fatigue explicitly.
- Use park + weather adjustments on totals and derivatives.
- Reprice instantly on lineup confirmations and starter scratches.
- Treat umpire and defense context as meaningful in totals/props.

8) What a High-Stakes Official Considers (Often Missed by Bettors)

- Who is betting (sharp/public profile), not just what is bet.
- Whether price moves are informational or liability-balancing.
- Correlation stacking risk inside SGP and derivative markets.

- Latency/stale feed exposure during lineup/injury windows.
- Market-by-market confidence and dynamic limits.
- Promo economics and effective hold dilution.
- Manual trader workflow/shift timing and operational blind spots.

9) Mirofish System Architecture Requirements

Data layer

- Multi-book odds snapshots with timestamps.
- Kalshi top-of-book and depth snapshots.
- Injury, lineup, starter/goalie, weather, travel/rest feeds.
- Closing lines for CLV benchmarking.

Feature layer

- No-vig market probabilities and cross-book dispersion.
- News-shock features and time-to-event decay.
- Liquidity and expected slippage proxies.
- League-specific context (for MLB: bullpen index, park factors).

Model layer

- Separate models by market type (moneyline/spread/total/props/binary events).
- Probability calibration (isotonic or Platt).
- Blend model output with market prior when appropriate.

Decision layer

- Trade only if edge exceeds dynamic threshold.
- Use fractional Kelly sizing with hard caps.
- Re-check edge at order time before execution.

10) Risk Policy

- Per-position max stake (% bankroll).
- Per-event and correlated exposure caps.
- Per-league and per-day exposure controls.
- Daily drawdown circuit breaker and cool-down logic.
- Stale-data and max-slippage abort conditions.

11) Performance and QA KPIs

Primary

- CLV

- ROI by market type
- Risk-adjusted return
- Max drawdown

Calibration

- Brier score
- Log loss
- Reliability curve drift

Execution

- Fill rate
- Average slippage
- Rejection rate by venue

12) Failure Modes and Red-Team Checklist

1. Overfitting short-term noise.
2. Ignoring fees and slippage.
3. Correlation blow-ups (especially props/SGP clusters).
4. Data latency during injury/lineup windows.
5. Model calibration drift not corrected in time.
6. Size escalation after losses.
7. Assuming liquidity exists at desired size.

Mitigations: freshness checks, automatic confidence decay, strict exposure graphing, continual recalibration, and hard risk circuit breakers.

13) Direct Mirofish Directive

For each candidate market, output:

1. Model probability with confidence interval.
2. Market implied probability (vig/fee adjusted).
3. EV after all explicit costs.
4. Stake recommendation (fractional Kelly + hard caps).
5. Correlation impact on existing open portfolio.
6. Abort decision if freshness/slippage/risk rules fail.

Optimization target: long-run risk-adjusted growth, not short-run win rate.

14) Sources Summary (for data context)

- Baseball-Reference: MLB 2025 major-league batting/pitching averages and year-by-year tables.

- Kalshi help/docs ecosystem for event contract and fee structure reference framing.

Note: Always re-validate current season stats and current venue fees before production use.